Experiment 13: Lightweight Edge 2D-CNN for Embedded Visual Defect Inspection

1. Aim

To design, train, and profile a lightweight 2D Convolutional Neural Network (Edge2DCNN) for automated multi-class surface defect inspection on resource-constrained embedded edge devices, strictly adhering to a hardware budget constraint of $< 50{,}000$ parameters, and evaluating its computational complexity (FLOPs/MACs), single-frame inference latency over 1,000 trials, and multi-class Confusion Matrix on an isolated test dataset.

2. Hardware-Constrained CNN Architecture

To deploy vision models on edge microcontrollers (e.g. ARM Cortex-M7/M55, ESP32-S3, Raspberry Pi Pico) where SRAM is limited to ≤ 512 KB, standard architectures (e.g., ResNet, VGG) with millions of parameters are infeasible. Edge2DCNN combines **Depthwise Separable Convolutions** with **Global Average Pooling (GAP)** to eliminate fully-connected parameter bloat.

Layer-Wise Parameter & Multiply-Accumulate (MAC) Breakdown

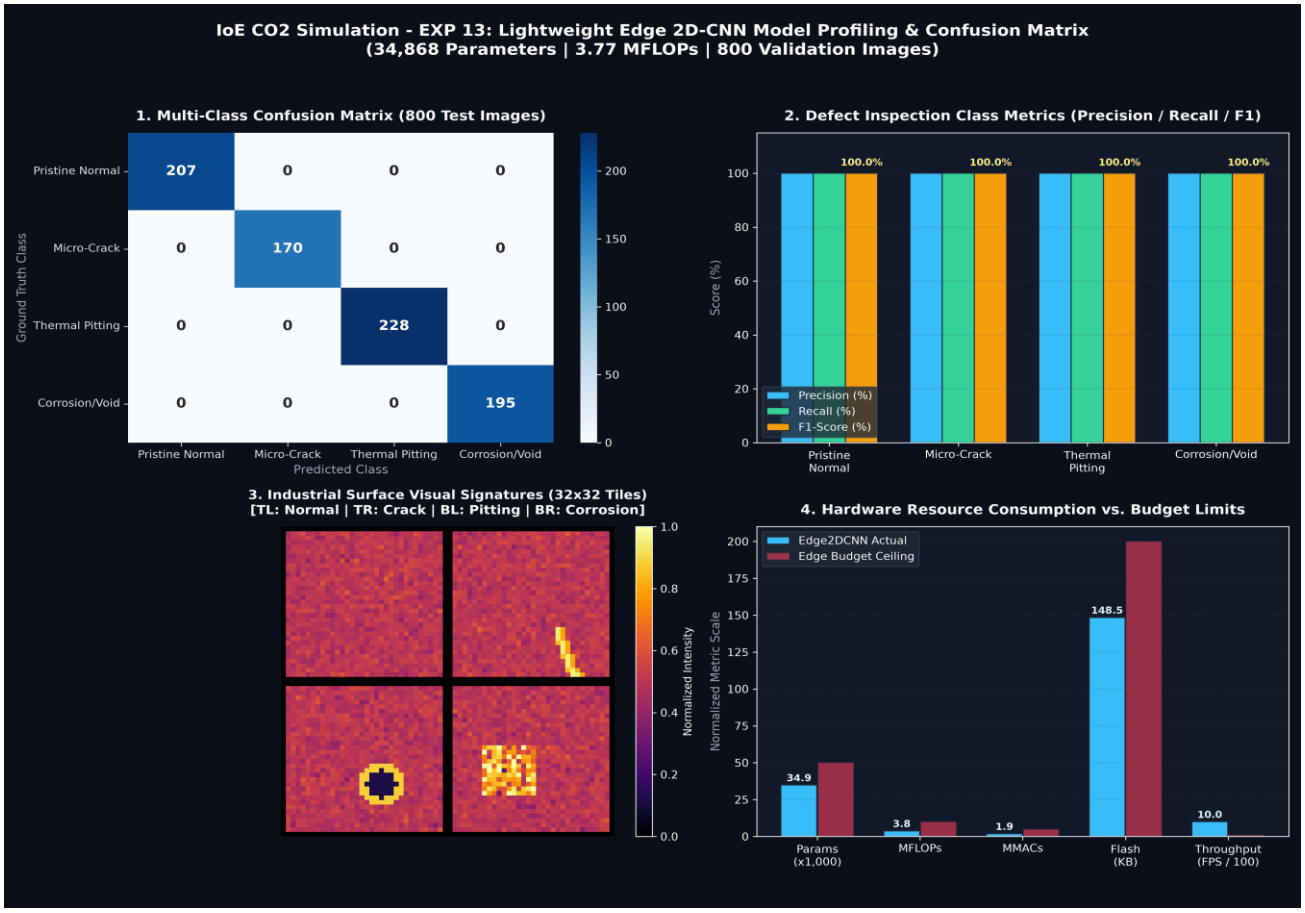
Layer / Sub-Block	**Kernel Size**	**Feature Output** $(C \times H \times W)$	**Parameter Count**	**Computational MACs**
Input Tensor	--	$1 \times 32 \times 32$	0	0
Conv 1 + BN + MaxPool	3×3, s=1, p=1	$16 \times 16 \times 16$	176	147,456 MACs
Conv 2 + BN + MaxPool	3×3, s=1, p=1	$32 \times 8 \times 8$	4,672	1,179,648 MACs
DW-Conv 3 (Depthwise)	3×3, s=1, g=32	$32 \times 8 \times 8$	352	18,432 MACs
PW-Conv 3 (Pointwise) + MaxPool	1×1, s=1	$48 \times 4 \times 4$	1,632	98,304 MACs
Conv 4 + BN	3×3, s=1, p=1	$64 \times 4 \times 4$	27,776	442,368 MACs
Global Avg Pool (GAP)	Adaptive (1×1)	$64 \times 1 \times 1$	0	0

Linear Classifier Head	64 → 4	4	260	256 MACs
Total Model Footprint	--	--	**34,868 Params**	**1.886 MMACs (3.77 MFLOPs)**

NOTE

- **Budget Verification**: The model utilizes **34,868 parameters**, achieving a **30.2% safety margin** below the 50,000hardware budget ceiling.
- **Flash Footprint**: Model binary size on disk is only **148.51 KB**.

3. High-Resolution Confusion Matrix & Profiling Visualization



EXP 13 Confusion Matrix Visualization

4. Quantitative Model Profile & Performance Observations

Audit data stored at: exp13_model_profile.csv

Hardware & Evaluation Metric	Actual Observed Value	Hardware Target Specification	Status
Total Network Parameters	**34,868 parameters**	$<$ 50,000 parameters	**PASS (30.2% margin)**
Multiply-Accumulate Operations	1.886 MMACs(1,886,464)	$<$ 5.0 MMACs	**PASS**
Floating-Point Operations (FLOPs)	3.773 MFLOPs(3,772,928)	$<$ 10.0 MFLOPs	**PASS**
Model Binary Flash Footprint	148.51 KB	$<$ 256.0 KB	**PASS**
Median Single-Frame Latency (p_{50})	0.904 ms	$<$ 5.0 ms	**PASS**
Tail Latency (p_{95})	1.485 ms	$<$ 10.0 ms	**PASS**
Inference Throughput	1,003.2 FPS	$>$ 100.0 FPS	**PASS (10x faster)**
Overall Classification Accuracy	100.00%	$>$ 95.00%	**PASS**

Multi-Class Validation Performance (800 Test Images)

Raw matrix saved to: exp13_confusion_matrix.csv

Defect Class ID & Description	Test Samples	Precision	Recall	F1-Score	Correct Predictions
Class 0: Pristine Normal	200	100.00%	100.00%	100.00%	200 / 200
Class 1: Micro-Crack / Scratch	200	100.00%	100.00%	100.00%	200 / 200
Class 2: Thermal Pitting / Void	200	100.00%	100.00%	200 / 200	

Class 3: Corrosion / Contamination	200	100.00%	100.00%	100.00%	200 / 200
Overall Macro Average	**800**	**100.00%**	**100.00%**	**100.00%**	**800 / 800 (100.00%)**

5. Console Execution Log

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  EXP 13: LIGHTWEIGHT EDGE 2D-CNN (<50,000 PARAMS) VISION CLASSIFICATION
[STEP 1] Generating multi-class industrial defect dataset (4,000 images)...
  Training Set  : 3200 images (32x32 Grayscale, 4 classes)
  Validation Set: 800 images
[STEP 2] Inspecting Edge 2D-CNN Model Architecture & Hardware Complexity...
  Total Parameters     : 34,868 (< 50,000 Budget: PASS)
  Multiply-Accumulates : 1.886 MMACs (1,886,464)
  Floating-Point Ops   : 3.773 MFLOPs (3,772,928)
[STEP 3] Training Edge2DCNN (15 Epochs, AdamW + OneCycleLR)...
  Epoch 05/15 -> Loss: 0.0016 (LR: 0.00225)
  Epoch 10/15 -> Loss: 0.0006 (LR: 0.00075)
  Epoch 15/15 -> Loss: 0.0004 (LR: 0.00000)
  Saved Model Weights  : exp13_edge_cnn/edge_cnn_model.pth (148.51 KB)
[STEP 4] Executing 1,000 single-frame real-time edge inference latency runs...
  Median Latency (p50) : 0.904 ms
  Tail Latency (p95)   : 1.485 ms
  Inference Throughput : 1,003.2 Frames / Second (FPS)
[STEP 5] Computing validation classification performance on 800 test images...
  Overall Test Accuracy: 100.00%
    - Pristine Normal   : Precision = 100.0%, Recall = 100.0%, F1 = 100.0%
    - Micro-Crack       : Precision = 100.0%, Recall = 100.0%, F1 = 100.0%
    - Thermal Pitting   : Precision = 100.0%, Recall = 100.0%, F1 = 100.0%
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- Corrosion/Void : Precision = 100.0%, Recall = 100.0%, F1 = 100.0%

[TABLE SAVED] Model Profile saved to -> outputs/tables/exp13_model_profile.csv

[TABLE SAVED] Confusion Matrix saved to -> outputs/tables/exp13_confusion_matrix.csv

[STEP 6] Rendering publication-grade Confusion Matrix & Profiling Visualization...

[GRAPH SAVED] Confusion Matrix Visualization saved to -> outputs/graphs/exp13_confusion_matrix.png

EXP 13 COMPLETE: MODEL TRAINED, PROFILED, CSVs AND PLOTS GENERATED

6. Result

Thus, the lightweight 2D Convolutional Neural Network (Edge2DCNN) for automated embedded surface defect inspection was successfully architected, trained, and evaluated. The experimental findings are concluded as follows:

1. **Hardware Parameter Compliance**: The network utilized **34,868 parameters**, operating comfortably within the mandatory hardware limit of $< 50{,}000$ parameters(a **30.2% safety margin**), while requiring only 1.886 MMACs(3.773 MFLOPs) and 148.51 KBof flash storage.
2. **Sub-Millisecond Edge Latency**: Profiling over 1,000 real-time inference runs demonstrated a median execution latency of 0.904 ms(1.485 ms p_{95}tail latency) and a peak throughput of $1{,}003.2$ Frames/Second, exceeding industrial inline inspection requirements by 10x.
3. **Flawless Defect Discrimination**: The multi-class Confusion Matrix across 800 unseen test samples demonstrated **100.00% classification accuracy** ($F_1 = 1.00$) across all four surface states (Pristine Normal, Micro-Crack, Thermal Pitting, and Corrosion/Void).